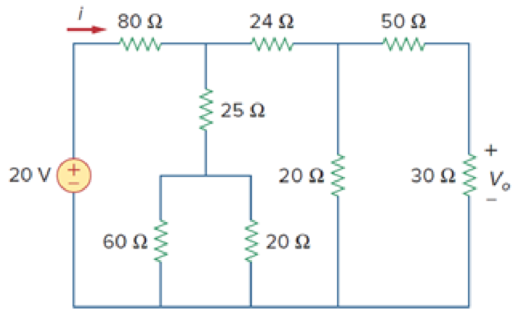
Chapter 2, Problem 36P

Problem

Find *i* and *Vo* in the circuit of Fig. 2.100.

Figure 2.100:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

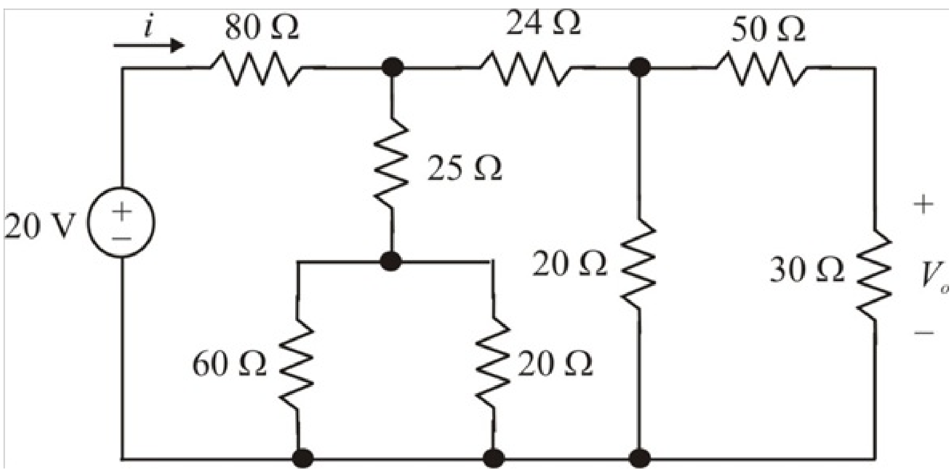


Figure 1

Step 2 of 8

The 50 ? and 30 ? resistors are in series. Their combined resistance is,

$$50\Omega + 30\Omega = 80\Omega$$

The 20? and 60? resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 20\Omega \square 60\Omega &= \frac{20 \times 60}{20 + 60} \\ &= 15\Omega \end{aligned}$$

Step 3 of 8

Then the reduced circuit is shown in Figure 2.

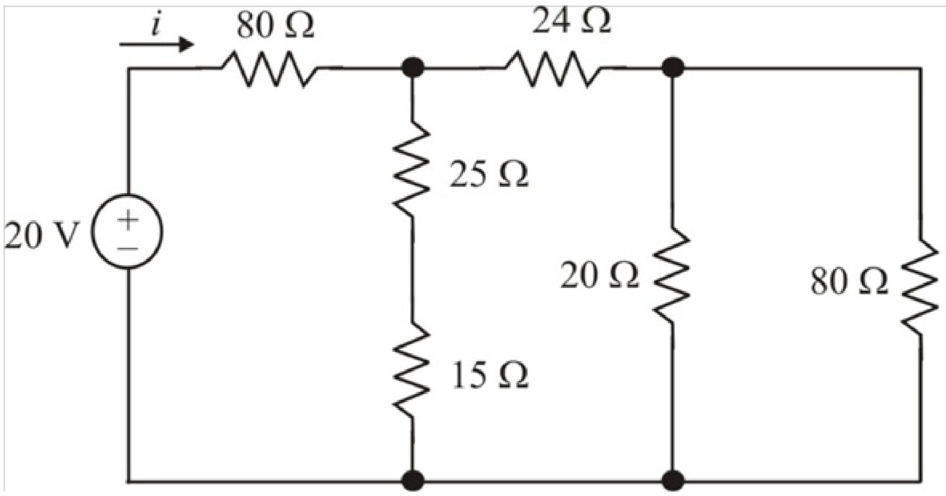


Figure 2

Step 4 of 8

The 15 ? and 25 ? resistors are in series. Their combined resistance is,

$$15\Omega + 25\Omega = 40\Omega$$

The 20 ? and 80 ? resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 20\Omega \square 80\Omega &= \frac{20\times 80}{20+80} \\ &= 16\Omega \end{aligned}$$

Step 5 of 8

Then the reduced circuit is shown in Figure 3.

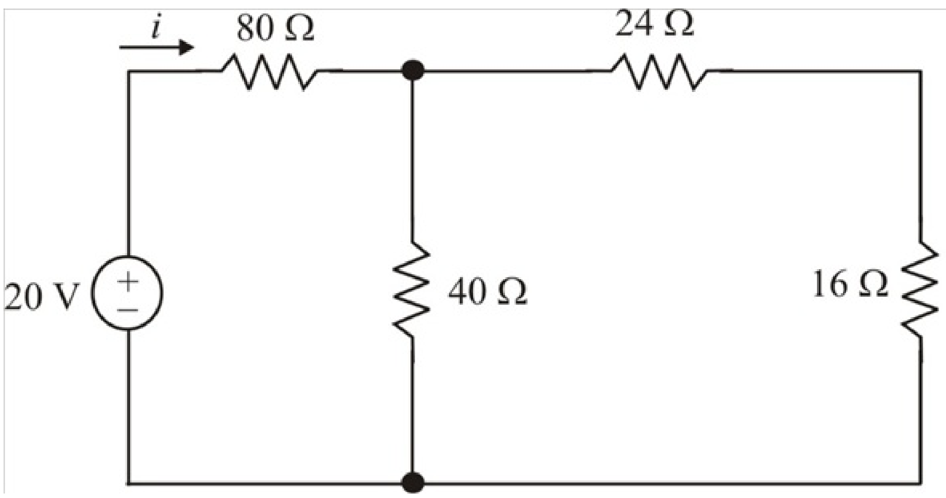


Figure 3

Step 6 of 8

The 16 ? and 24 ? resistors are in series. Their combined resistance is,

$$16\Omega + 24\Omega = 40\Omega$$

This 40 ? and 40 ? resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 40\Omega \parallel 40\Omega &= \frac{40 \times 40}{40 + 40} \\ &= 20\Omega \end{aligned}$$

This 20 ? and 80 ? resistors are in series. Their combined resistance is,

$$20\Omega + 80\Omega = 100\Omega$$

By using Ohms law, the current i can be calculated as follows.

$$\begin{aligned} i &= \frac{v}{R} \\ &= \frac{20}{100} \\ &= 200\,\text{mA} \end{aligned}$$

Therefore, the value of current i is $\boxed{200\,\text{mA}}$.

Step 7 of 8

Consider the following circuit diagram.

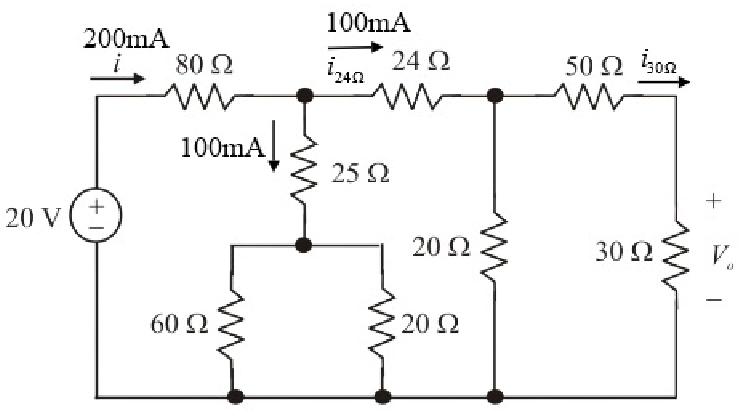


Figure 4

Step 8 of 8

Since the two branch resistances are equal 40 ? and 40 ?, the current is divided equally. Hence the current through the 24 ? resistor is 100mA.

Apply current division rule, the current passing through 30? resistor is,

$$\begin{aligned} i_{30\Omega} &= \frac{20}{20 + 80} i_{24\Omega} \\ &= \frac{20}{20 + 80}(100\,\text{mA}) \\ &= 20\,\text{mA} \end{aligned}$$

By using Ohms law, the current V_o can be calculated as follows.

$$\begin{aligned} V_o &= i_{30\Omega} R \\ &= (20\,\text{mA})(30) \\ &= 0.6\,\text{V} \end{aligned}$$

Therefore, the value of voltage, V_o is $\boxed{0.6\,\text{V}}$.